

Robotic Microscope Power Distribution
Detailed Power Routing and Current Paths

CRITICAL NOTES

Grounding Strategy

- Single-point ground at bus bar
 - Prevents ground loops
 - All GND wires meet at bus
- No chassis ground connection

Fusing

- 5A automotive blade fuse
 - Protects against shorts
 - Located at 24V input

Connectors (External)

- All external: JST-SM series
 - Motors: 4-pin JST
 - Limits: 2-pin JST
 - LED: 3-pin JST
 - eStop: 2-pin JST (future)

Safety

⚠ 24V is considered SELV

⚠ Fuse is essential

⚠ Insulate all connections

⚠ No exposed conductors

POWER BUDGET

System Power Consumption

Component	Current	Power
Teensy 4.1	150mA @ 5V	0.75W
RPi4	2.5A @ 5V	12.5W
LED Ring (40%)	500mA @ 5V	2.5W
Level Shifter	10mA @ 5V	0.05W
5V Subtotal	3.16A	15.8W
Motors (4× 1A)	4A @ 24V	96W
24V Subtotal	4A	96W
Total System	~4A @ 24V	~112W

Supply: 24V 5A = 120W
✓ Excellent headroom (8W margin)
TB6600s configured for 1A via DIP switches

AC INPUT

120V AC Wall Outlet

AC Power

AC/DC Adapter
24V 5A
(120W Max)

